Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Contents

Integrals

1.1 Medium Integral

Absolute Value Exponential Integral

Evaluate the definite integral:

$$\int_{-\infty}^{\infty} \left| x^3 2^{-x^2} \right| dx$$

Solution: Let the integral be denoted by I. We first note that the integrand $f(x) = \left| x^3 2^{-x^2} \right|$ is an even function. This is because:

$$f(-x) = \left| (-x)^3 2^{-(-x)^2} \right| = \left| -x^3 \right| 2^{-x^2} = \left| x^3 \right| 2^{-x^2} = f(x)$$

Because the integrand is symmetric about the y-axis, we can evaluate the integral from 0 to ∞ and multiply the result by 2. For $x \geq 0$, $x^3 \geq 0$, so we can drop the absolute value bars:

$$I = 2 \int_0^{\infty} x^3 2^{-x^2} \, dx$$

To evaluate this, we use the substitution $u = x^2$. Then, $du = 2x \, dx$. We can rewrite the integrand as:

$$I = \int_0^{\infty} (x^2) 2^{-x^2} (2x \, dx) = \int_0^{\infty} u 2^{-u} \, du$$

This is a standard integral that can be evaluated using integration by parts. Let $w = u$ and $dv = 2^{-u} \, du$. Then, $dw = du$ and $v = -\frac{2^{-u}}{\ln(2)}$. Applying the integration by parts formula $\int w \, dv = wv - \int v \, dw$:

$$I = \left[-\frac{u 2^{-u}}{\ln(2)} \right]_0^{\infty} - \int_0^{\infty} \left(-\frac{2^{-u}}{\ln(2)} \right) du$$

Evaluating the boundary term: As $u \to \infty$, $u 2^{-u} = \frac{u}{2^u} \to 0$ (exponential growth dominates polynomial). At $u = 0$, the term is exactly 0. Thus, the boundary term vanishes.

$$I = 0 + \frac{1}{\ln(2)} \int_0^{\infty} 2^{-u} \, du$$

Integrating the remaining exponential function:

$$I = \frac{1}{\ln(2)} \left[-\frac{2^{-u}}{\ln(2)} \right]_0^{\infty} = \frac{1}{\ln(2)} \left(0 - \left(-\frac{1}{\ln(2)} \right) \right) = \frac{1}{(\ln(2))^2}$$

The exact value is:

$$\frac{1}{(\ln 2)^2}$$

1.2 Hard Integral

Rational Exponential Double Integral

Evaluate the definite integral:

$$\int_0^1 \int_0^{1-x} e^{\left(\frac{y}{x+y}\right)}\, dy\, dx$$

Solution: Let the target integral be denoted by I. The domain of integration D is a triangle defined by $0 \leq x \leq 1$ and $0 \leq y \leq 1-x$, which implies $x+y \leq 1$ along with $x, y \geq 0$. The presence of $x+y$ in the denominator of the exponent suggests a transformation of variables. We apply the standard substitution:

$$u = x + y, \quad v = \frac{y}{x+y}$$

From these definitions, we can express x and y in terms of u and v:

$$y = uv, \quad x = u - y = u(1-v)$$

Next, we compute the Jacobian determinant of the transformation, $J = \frac{\partial(x,y)}{\partial(u,v)}$:

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1-v & -u \\ v & u \end{vmatrix}$$

$$J = (1-v)(u) - (-u)(v) = u - uv + uv = u$$

Now we determine the new limits of integration. Since $x, y \geq 0$, we have $u \geq 0$ and $0 \leq v \leq 1$. The boundary $x + y \leq 1$ translates directly to $u \leq 1$. Thus, the new region of integration is a simple rectangle in the uv-plane: $0 \leq u \leq 1$ and $0 \leq v \leq 1$. Applying the transformation to the integral:

$$I = \int_0^1 \int_0^1 e^v |J|\, dv\, du = \int_0^1 \int_0^1 e^v u\, dv\, du$$

Because the limits are constants and the integrand factors into independent functions of u and v, we can separate the integral into a product of two single-variable integrals:

$$I = \left(\int_0^1 u\, du \right) \left(\int_0^1 e^v\, dv \right)$$

Evaluating each integral independently:

$$\int_0^1 u\, du = \left[\frac{u^2}{2} \right]_0^1 = \frac{1}{2}$$

$$\int_0^1 e^v\, dv = [e^v]_0^1 = e - 1$$

Multiplying the results yields the final exact value:

$$I = \frac{1}{2}(e-1)$$

Differentials

2.1 Medium Differential

Riccati / Homogeneous Differential Equation

Solve the differential equation:

$$x\frac{dy}{dx} - y = x^2 + y^2$$

Given the initial condition $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, find the exact value of $y\left(\frac{\pi}{4}\right)$.

Solution: We rewrite the given first-order nonlinear differential equation by dividing by x^2:

$$\frac{1}{x}\frac{dy}{dx} - \frac{y}{x^2} = 1 + \left(\frac{y}{x}\right)^2$$

This equation is heavily suggestive of the substitution $y = ux$. Differentiating with respect to x using the product rule gives:

$$\frac{dy}{dx} = u + x\frac{du}{dx}$$

Substituting y and $\frac{dy}{dx}$ back into the original equation:

$$x\left(u + x\frac{du}{dx}\right) - ux = x^2 + (ux)^2$$

Expanding and simplifying the left side:

$$ux + x^2\frac{du}{dx} - ux = x^2 + u^2x^2 \implies x^2\frac{du}{dx} = x^2(1+u^2)$$

Since $x \neq 0$ for our domain of interest, we can divide both sides by x^2:

$$\frac{du}{dx} = 1 + u^2$$

This is a separable differential equation. We isolate u and x:

$$\frac{du}{1+u^2} = dx$$

Integrating both sides:

$$\int \frac{du}{1+u^2} = \int dx \implies \arctan(u) = x + C$$

Applying the tangent function to solve for u:

$$u(x) = \tan(x + C)$$

Substituting back $u = \frac{y}{x}$, the general solution to the differential equation is:

$$y(x) = x \tan(x + C)$$

We use the initial condition $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$ to find the constant C:

$$\frac{\pi}{2} = \frac{\pi}{2} \tan\left(\frac{\pi}{2} + C\right)$$

Dividing by $\frac{\pi}{2}$:

$$1 = \tan\left(\frac{\pi}{2} + C\right)$$

The tangent function equals 1 at $\frac{\pi}{4} + n\pi$. Thus:

$$\frac{\pi}{2} + C = \frac{\pi}{4} \implies C = \frac{\pi}{4} - \frac{\pi}{2} = -\frac{\pi}{4}$$

The particular solution is therefore:

$$y(x) = x \tan\left(x - \frac{\pi}{4}\right)$$

Finally, we evaluate this function at $x = \frac{\pi}{4}$:

$$y\left(\frac{\pi}{4}\right) = \frac{\pi}{4} \tan\left(\frac{\pi}{4} - \frac{\pi}{4}\right) = \frac{\pi}{4} \tan(0)$$

Since $\tan(0) = 0$:

$$y\left(\frac{\pi}{4}\right) = 0$$

2.2 Hard Differential

First-Order Linear Differential Equation

Given the first-order linear differential equation:

$$y' - \frac{3y}{x} = x^2 \ln^3(1 - x)$$

Find the constant of integration C such that $y\left(\frac{1}{2}\right) = \frac{\zeta(3)}{2}$.

Solution: This is a first-order linear ordinary differential equation of the form $y' + P(x)y = Q(x)$, where $P(x) = -\frac{3}{x}$. We begin by finding the integrating factor $\mu(x)$:

$$\mu(x) = \exp\left(\int -\frac{3}{x}\, dx\right) = \exp(-3 \ln x) = x^{-3}$$

Multiplying both sides of the differential equation by the integrating factor gives:

$$x^{-3}y' - 3x^{-4}y = x^{-1} \ln^3(1 - x)$$

The left side is exactly the derivative of the product $(x^{-3}y)$:

$$\frac{d}{dx}\left(x^{-3}y\right) = \frac{\ln^3(1-x)}{x}$$

Integrating both sides with respect to x, we construct the general solution with the lower bound of integration set to 0 to cleanly capture the indefinite constant:

$$x^{-3}y = \int_0^x \frac{\ln^3(1-t)}{t}\,dt + C$$

Multiplying by x^3 isolates $y(x)$:

$$y(x) = x^3\left(\int_0^x \frac{\ln^3(1-t)}{t}\,dt + C\right)$$

We are given the initial condition $y\left(\frac{1}{2}\right) = \frac{\zeta(3)}{2}$. Substituting $x = \frac{1}{2}$ into our equation:

$$y\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^3\left(\int_0^{1/2} \frac{\ln^3(1-t)}{t}\,dt + C\right) = \frac{1}{8}\left(\int_0^{1/2} \frac{\ln^3(1-t)}{t}\,dt + C\right)$$

Setting this equal to the given condition:

$$\frac{1}{8}\left(\int_0^{1/2} \frac{\ln^3(1-t)}{t}\,dt + C\right) = \frac{\zeta(3)}{2}$$

Multiplying both sides by 8 yields:

$$\int_0^{1/2} \frac{\ln^3(1-t)}{t}\,dt + C = 4\zeta(3)$$

Isolating C, the constant of integration is explicitly determined by the definite integral evaluation:

$$C = 4\zeta(3) - \int_0^{1/2} \frac{\ln^3(1-t)}{t}\,dt$$

Derivatives

3.1 Medium Derivative

Infinite Nested Radical Derivative

Let $a_1 = \sqrt{x}$ and $a_t = \sqrt{x + a_{t-1}}$ for $t > 1$. If $y = \lim_{n\to\infty} a_n$, find $y''(2)$.

Solution: We are given an infinite nested sequence a_n. Assuming the limit converges to y, we can write the recursive relationship at the limit as:

$$y = \sqrt{x + y}$$

To find the derivatives of y with respect to x, we first eliminate the radical by squaring both sides:

$$y^2 = x + y$$

Rearranging into a standard quadratic form in terms of y:

$$y^2 - y - x = 0$$

Using the quadratic formula to solve for y:

$$y = \frac{1 \pm \sqrt{(-1)^2 - 4(1)(-x)}}{2} = \frac{1 \pm \sqrt{1+4x}}{2}$$

Since $a_1 = \sqrt{x} > 0$ for $x > 0$, the sequence consists strictly of positive terms, meaning y must be positive. We therefore take the positive root:

$$y(x) = \frac{1}{2}\left(1 + \sqrt{1+4x}\right) = \frac{1}{2} + \frac{1}{2}(1+4x)^{1/2}$$

Now we find the first derivative $y'(x)$ using the chain rule:

$$y'(x) = \frac{1}{2} \cdot \frac{1}{2}(1+4x)^{-1/2} \cdot \frac{d}{dx}(1+4x)$$

$$y'(x) = \frac{1}{4}(1+4x)^{-1/2} \cdot 4 = (1+4x)^{-1/2} = \frac{1}{\sqrt{1+4x}}$$

Next, we differentiate $y'(x)$ to find the second derivative $y''(x)$:

$$y''(x) = -\frac{1}{2}(1+4x)^{-3/2} \cdot 4 = -2(1+4x)^{-3/2} = \frac{-2}{(1+4x)^{3/2}}$$

Finally, we evaluate the second derivative at $x = 2$:

$$y''(2) = \frac{-2}{(1+4(2))^{3/2}} = \frac{-2}{(1+8)^{3/2}} = \frac{-2}{9^{3/2}}$$

Since $9^{3/2} = (\sqrt{9})^3 = 3^3 = 27$:

$$y''(2) = -\frac{2}{27}$$

3.2 Hard Derivative

Roots of Unity Product Derivative

Let $f(x) = \prod_{k=1}^{2026} \left(x^2 - 2x\cos\left(\frac{k\pi}{2027}\right) + 1 \right)$. Find $f'(1)$.

Solution: Let $n = 2027$. The function is a product of $n - 1$ quadratic factors. We can recognize these factors by exploring the factorization of $x^{2n} - 1$. The complex roots of $x^{2n} - 1 = 0$ are the $2n$-th roots of unity, given by $x = e^{i\frac{k\pi}{n}}$ for $k = 0, 1, 2, \dots, 2n-1$. The roots for $k = 0$ and $k = n$ are real: $x = 1$ and $x = -1$, which correspond to the polynomial $(x-1)(x+1) = x^2 - 1$. The remaining $2n - 2$ complex roots appear in conjugate pairs: $e^{i\frac{k\pi}{n}}$ and $e^{-i\frac{k\pi}{n}}$ for $k = 1, 2, \dots, n-1$. Multiplying a conjugate pair of linear factors yields a real quadratic factor:

$$\left(x - e^{i\frac{k\pi}{n}}\right)\left(x - e^{-i\frac{k\pi}{n}}\right) = x^2 - x\left(e^{i\frac{k\pi}{n}} + e^{-i\frac{k\pi}{n}}\right) + 1 = x^2 - 2x\cos\left(\frac{k\pi}{n}\right) + 1$$

Therefore, the polynomial $x^{2n} - 1$ can be completely factored over the reals as:

$$x^{2n} - 1 = (x^2 - 1)\prod_{k=1}^{n-1}\left(x^2 - 2x\cos\left(\frac{k\pi}{n}\right) + 1\right)$$

Our function $f(x)$ is exactly this product with $n = 2027$. We can thus express $f(x)$ elegantly as:

$$f(x) = \frac{x^{2n} - 1}{x^2 - 1}$$

Recognizing this as the sum of a finite geometric series with common ratio x^2, we rewrite $f(x)$ as a polynomial:

$$f(x) = \sum_{j=0}^{n-1} (x^2)^j = 1 + x^2 + x^4 + \cdots + x^{2n-2}$$

We now differentiate this polynomial term-by-term:

$$f'(x) = 2x + 4x^3 + 6x^5 + \cdots + (2n-2)x^{2n-3} = \sum_{j=1}^{n-1} 2jx^{2j-1}$$

We evaluate the derivative at $x = 1$. Since 1 raised to any power is 1, this simply becomes the sum of an arithmetic progression:

$$f'(1) = \sum_{j=1}^{n-1} 2j = 2\sum_{j=1}^{n-1} j = 2\left(\frac{(n-1)n}{2}\right) = n(n-1)$$

Substituting $n = 2027$ into our resulting formula:

$$f'(1) = 2027 \times 2026$$

Calculating the exact numerical product:

$$f'(1) = 4,106,702$$

Physics & Engineering

4.1 Mechanics

Kepler's Third Law in a Binary System

The distance between two stars of masses $3M_S$ and $6M_S$ is $9R$. Here R is the mean distance between the centers of the Earth and the Sun, and M_S is the mass of the Sun. The two stars orbit around their common center of mass in circular orbits with period nT, where T is the period of Earth's revolution around the Sun. Find the value of n.

Solution: For a binary star system orbiting a common center of mass, the generalized form of Kepler's Third Law applies. The period T_{bin} of the binary system is given by:

$$T_{bin}^2 = \frac{4\pi^2}{G(M_1 + M_2)} a^3$$

where G is the gravitational constant, M_1 and M_2 are the stellar masses, and a is the constant orbital separation distance. In our binary system: - Total mass $M_1 + M_2 = 3M_S + 6M_S = 9M_S$ - Orbital separation $a = 9R$ Substituting these into the formula:

$$T_{bin}^2 = \frac{4\pi^2}{G(9M_S)}(9R)^3 = \frac{4\pi^2}{9GM_S}(729R^3)$$

Simplifying the expression by factoring out 9 from 729:

$$T_{bin}^2 = 81\left(\frac{4\pi^2 R^3}{GM_S}\right)$$

Now, consider the Earth-Sun system. The mass of the Earth is negligible compared to the Sun ($M_E \ll M_S$), so the total mass is approximately M_S. The semi-major axis is R. Kepler's Third Law for Earth is:

$$T^2 = \frac{4\pi^2 R^3}{GM_S}$$

Substituting T^2 back into the equation for the binary system:

$$T_{bin}^2 = 81T^2$$

Taking the square root of both sides gives the period of the binary system in terms of Earth's orbital period:

$$T_{bin} = 9T$$

Since the problem states the period is nT, we determine the coefficient:

$$n = 9$$

4.2 Quantum Mechanics

Hydrogenic Ion Energy Transitions

The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light, the ion undergoes a transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom. What is the energy of the state S_1 in units of the hydrogen atom ground state energy?

Solution: The energy levels of a hydrogen-like ion are given by the Bohr formula:

$$E_n = Z^2 \frac{E_H}{n^2}$$

where Z is the atomic number, n is the principal quantum number, and E_H is the ground state energy of the hydrogen atom (-13.6 eV). For Li^{2+}, $Z = 3$. The number of radial nodes for a hydrogenic orbital is given by the formula $n - l - 1$, where l is the azimuthal (angular momentum) quantum number. Let's analyze state S_1: Because S_1 is "spherically symmetric", it is an s-orbital, meaning $l = 0$. We are given that it has exactly 1 radial node. Using the node formula:

$$n_1 - 0 - 1 = 1 \implies n_1 = 2$$

Thus, S_1 corresponds to the $2s$ state of Li^{2+}. We can calculate its energy in terms of E_H:

$$E_{S1} = 3^2 \frac{E_H}{2^2} = \frac{9}{4} E_H = 2.25 E_H$$

To verify consistency, let's analyze state S_2: The energy of S_2 is equal to the ground state energy of Hydrogen (E_H).

$$E_{S2} = 3^2 \frac{E_H}{n_2^2} = E_H \implies \frac{9}{n_2^2} = 1 \implies n_2 = 3$$

State S_2 is in the $n = 3$ shell. We are told S_2 has exactly 1 radial node. Using the node formula again:

$$3 - l_2 - 1 = 1 \implies 2 - l_2 = 1 \implies l_2 = 1$$

So S_2 is a $3p$ orbital. The transition from $2s$ to $3p$ perfectly satisfies the dipole selection rule ($\Delta l = \pm 1$) for absorbing a photon. The question asks for the energy of state S_1 in units of the hydrogen atom ground state energy, which is strictly the coefficient:

$$2.25$$

4.3 Electromagnetism

Motional EMF in a Magnetic Field

A hamster is fired horizontally from a mangonel off the roof of a tall building. He is equipped with a parachute connected to a straight length of copper wire folded over itself to create a complete circuit. If a current of at least $I = 1$ mA flows the parachute opens. The wire has a diameter $d = 2$ mm and resistivity $\rho = 1.7 \times 10^{-8}\,\Omega$m. The hamster is launched through a magnetic field of $B = 500$ nT. Assuming the wire remains perpendicular to the field and the motion, find the minimum launch velocity v (in m s^{-1}) for safety. Give your answer to 2 significant figures.

Solution: Let the total length of the copper wire be L. Because it is "folded over itself to create a complete circuit", it acts like a U-shaped loop (or a pair of parallel rails connected at the ends). When this loop travels perpendicularly through a magnetic field boundary, only the leading folded edge of length $L/2$ actively cuts the magnetic flux to generate a motional electromotive force (EMF). The motional EMF ϵ generated is:

$$\epsilon = B\left(\frac{L}{2}\right)v$$

The electrical resistance R of the entire length L of wire is determined by its resistivity ρ, length L, and cross-sectional area $A = \pi\left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$:

$$R = \rho\frac{L}{A} = \frac{4\rho L}{\pi d^2}$$

According to Ohm's Law, the current I induced in the circuit is $I = \frac{\epsilon}{R}$. Substituting our expressions:

$$I = \frac{B\left(\frac{L}{2}\right)v}{\frac{4\rho L}{\pi d^2}} = \frac{BLv\pi d^2}{8\rho L}$$

Notice that the arbitrary length of the wire L cleanly cancels out, leaving:

$$I = \frac{\pi B v d^2}{8\rho}$$

We rearrange the equation to solve for the launch velocity v:

$$v = \frac{8\rho I}{\pi d^2 B}$$

Now we convert all given parameters to standard SI units: - Resistivity $\rho = 1.7 \times 10^{-8}\,\Omega$m - Current threshold $I = 1$ mA $= 10^{-3}$ A - Wire diameter $d = 2$ mm $= 2 \times 10^{-3}$ m $\implies d^2 = 4 \times 10^{-6}$ m^2 - Magnetic field $B = 500$ nT $= 500 \times 10^{-9}$ T $= 5 \times 10^{-7}$ T Substituting the values into the velocity equation:

$$v = \frac{8 \times (1.7 \times 10^{-8}) \times 10^{-3}}{\pi \times (4 \times 10^{-6}) \times (5 \times 10^{-7})}$$

Simplifying the numerator and denominator:

$$v = \frac{13.6 \times 10^{-11}}{\pi \times 20 \times 10^{-13}} = \frac{13.6 \times 10^{-11}}{20\pi \times 10^{-13}} = \frac{13.6}{20\pi} \times 10^2 = \frac{1360}{20\pi} = \frac{68}{\pi}$$

Approximating $\pi \approx 3.14159$, we find the numerical value:

$$v \approx \frac{68}{3.14159} \approx 21.645 \text{ m s}^{-1}$$

Rounding to two significant figures as requested:

$$22$$